

wider frequency range, and have minimal training requirements.

Non-invasive BCIs

Non-invasive BCI sensors are placed on or very close to the head to measure the brain activity, thus avoiding any surgery. EEGs are the most commonly used ones. Another mechanism used is magnetoencephalography (MEG)—a non-invasive neurophysiological technique that measures the magnetic fields generated by neuronal activity of the brain. It is a direct measure of brain function providing a very high temporal resolution.

Non-invasive devices are easy to wear, have lower costs and greater portability. They have relatively poor spatial resolution and are unable to effectively use higher frequencies as the skull dampens the signals, dispersing and blurring the electromagnetic waves created by the neurons. Compared to invasive devices, these are susceptible to noise and have worse signal resolution due to the distance from the brain.



Fig. 5: A non-invasive BCI (Source: <http://learn.neurotechedu.com/introtobci>)

BCI TECHNOLOGIES

BCIs based on visual evoked potential

Visual evoked potentials (VEPs) are the brain activity modulations occurring in the visual cortex after encountering visual stimulus. They are easy to detect as the movement of stimulus closer to the central visual field greatly enhances the amplitude of the VEPs. In VEP based BCIs, the electrical potentials caused by visual stimuli such as an alternating checkerboard pattern or a flashing light are recorded on an EEG using electrodes placed at the back of head.

VEP based BCIs are becoming popular given the advantage of less user training that is required and high information transfer rate (ITR). VEP based BCIs include steady-state visually evoked potential (SSVEP) BCIs and P300 event related potential (ERP) based BCIs.

In SSVEP based BCI systems, when the eyes are excited by a visual stimulus signal, the brain generates response at the same frequency. The best responses are obtained for frequencies between 5 and 20Hz. The response signals of these stimuli are detected from the scalp and analysed. Stimuli used are a periodic visual stimulation such as alternating checkerboard patterns or flashing images.

The signals elicited have high signal-to-noise ratio, are robust, and are easy to record. However, these may be irritating and uncomfortable to the users, as they have to gaze for long at

the flashing light or iterating symbols during the sessions. SSVEP based BCIs can provide hands-free human interaction with the environment.

Another type of VEP based BCI used is the P300VEP—a positive peak in the EEG that occurs at roughly 300ms after the appearance of a target stimulus, which the user is seeking or oddball stimuli. The P300 amplitude decreases as the target stimuli and the ignored stimuli become more similar.

In an experiment with P300 based BCI, the subject was asked to choose a letter from a grid of 6 by 6 letters and numbers, the EEG signals from the subject's scalp were measured and processed in real time to detect the P300 potential, which reflected the subject's imagined choice letter. The advantage of P300 is that the subject has to undertake shorter training to learn the use of the BCI paradigm

BCIs based on electrooculography

The technique of measuring cornea retinal potential associated with eye movement is called electrooculography (EOG). An eye is an electric dipole. Changes in the position of the eyeball produce variations in the electric charge registered by the electrodes.

EOG is used for measuring the corneo-retinal standing potential that exists between the front and the back of a human eye and the resulting signal, called electrooculogram, is used for recording eye movements to provide the sensitive visual stimuli.

In an experiment, a robot could be driven from start to a goal point by EOG based commands—forward, backward, left, right, and stop, using the visual fixation by the subject. A human-computer interface has also been developed, which does not require visual fixation or movement of the eyes and is based on directing attention to a chosen letter on a virtual keyboard by looking directly at the letter.

BCIs based on magnetoencephalography and functional magnetic resonance imaging (fMRI)

Magnetoencephalography (MEG) is a functional neuroimaging technique for

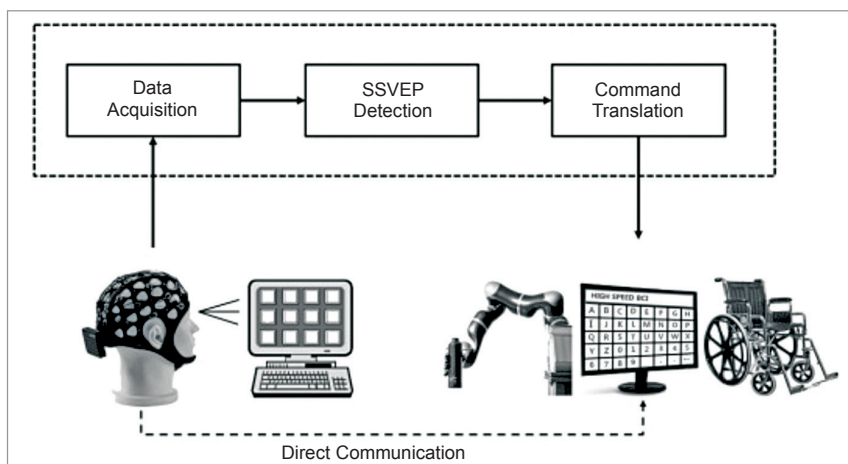


Fig. 6: SSVEP based BCI (Source: www.sciencedirect.com)